

ENTERPRISE SECURITY ASSESSMENT LAB

AD NETWORK SECURITY METHODOLOGY REPORT

Phase 7 - Visualizing Active Directory privilege relationships

Report Section	Phase 07 - BloodHound Attack Path Analysis
Project Area	Internal Network and Active Directory Security Assessment
Environment	Private Windows AD lab - Kali Linux, Windows Server, Windows endpoint
Prepared For	GitHub Portfolio Documentation
Document Type	Individual methodology PDF with redacted evidence

Authorized lab documentation only. Sensitive credential material, hashes, passwords, and replication output have been redacted before publication.

1. Executive Summary

This report documents **BloodHound Attack Path Analysis** as part of the AD Network module in the Enterprise Security Assessment Lab. The section is written as a professional, portfolio-safe methodology report that explains the testing objective, workflow, evidence, tool usage, sanitized output, security risk, remediation guidance, and GitHub publication requirements.

Objective: Collect and visualize Active Directory relationships to identify privilege paths, group relationships, and possible routes from owned principals to higher privileges.

MITRE / Framework Mapping	T1069.002 - Domain Groups; T1482 - Domain Trust Discovery; defensive attack-path analysis
Testing Scope	Authorized private Active Directory lab only
Evidence Handling	Screenshots embedded with necessary redaction; raw hashes and secrets are not included

2. Methodology and Workflow

The methodology below is designed to show a controlled penetration-testing workflow rather than random tool execution. Each step supports a specific assessment goal and produces evidence that can be reviewed safely in a public portfolio.

- Start Neo4j and BloodHound in the test environment.
- Collect AD relationship data using authorized credentials.
- Import the collection ZIP into BloodHound.
- Run path queries such as shortest paths to Domain Admins and domain overview.
- Document only sanitized screenshots that do not expose unnecessary usernames or hashes.

Command / Workflow Used

```
# Start Neo4j
sudo neo4j start

# Start BloodHound GUI
bloodhound

# Collect data using BloodHound Python collector
bloodhound-python \
-d corp.local \
-u <REDACTED_USER> \
-p '<REDACTED_PASSWORD>' \
-dc <DC_HOSTNAME_OR_IP> \
-c all \
--zip
```

Sanitized Output Style

```
Collector output: <TIMESTAMP>_bloodhound.zip
Analysis views:
- Shortest paths to Domain Admins
- Owned principals
- Group memberships
- Local admin relationships
- Domain overview graph
```

3. Evidence Mapping and Screenshot Explanation

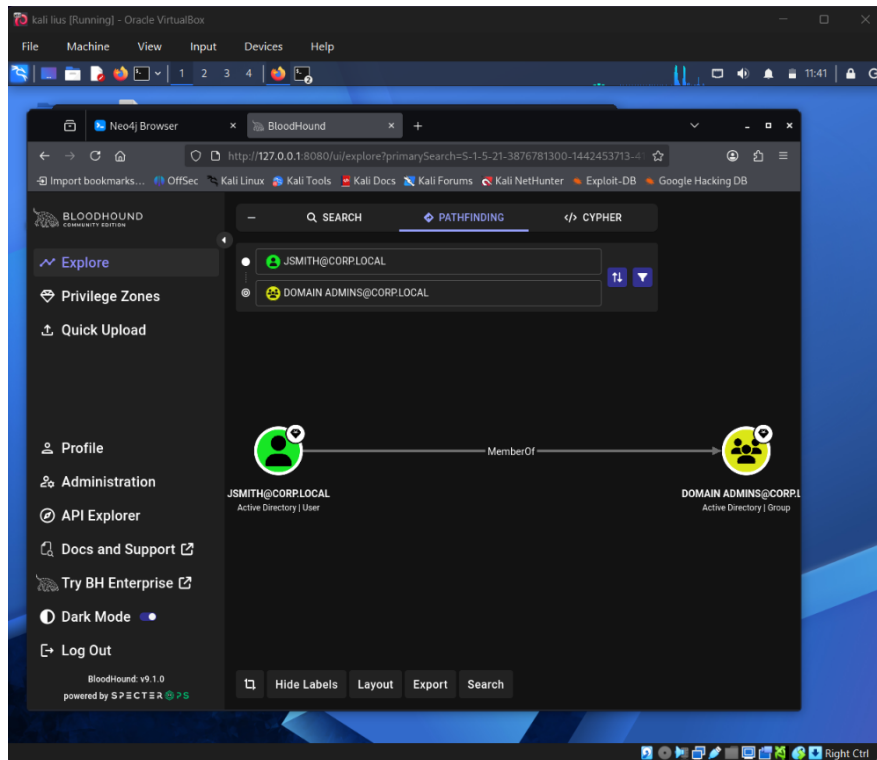


Figure - BloodHound shortest path to Domain Admin

This screenshot visualizes a lab attack path toward high privilege.

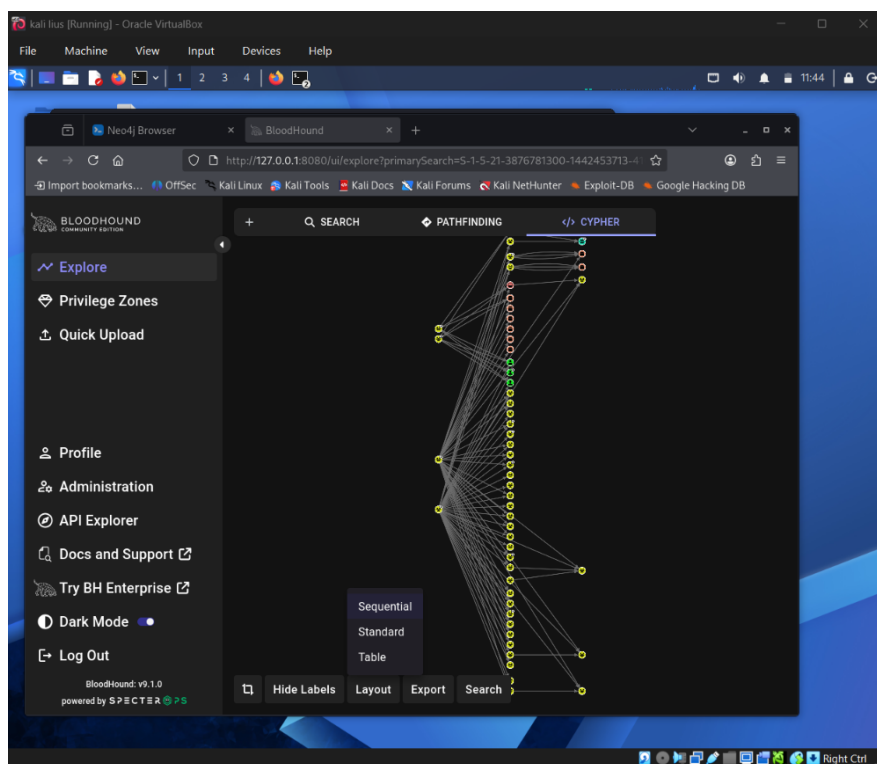


Figure - BloodHound domain overview graph

This screenshot shows broader domain relationship mapping and graph-based AD analysis.

4. Technical Interpretation

The evidence is interpreted in the context of a private AD lab. The screenshot proves execution of the relevant technique or validation step, while the sanitized command and output blocks show how the result should be presented without exposing raw credential material.

What the evidence proves	The lab phase for BloodHound Attack Path Analysis was executed and documented.
What it does not prove	It does not indicate testing of any public, third-party, or production network.
Publication status	Safe for GitHub after redaction of sensitive values.

5. Risk Analysis

BloodHound-style analysis often reveals privilege relationships that are difficult to identify manually. Misconfigured group membership, delegated rights, local admin paths, and excessive privileges can combine into a practical route to domain compromise.

Real-World Impact Categories

- Credential exposure or credential reuse risk
- Lateral movement or privilege escalation potential depending on account rights
- Domain visibility and attack path discovery in internal environments
- Operational and detection gaps if monitoring does not cover the relevant Windows, Kerberos, SMB, or LDAP events

6. Recommended Remediation

- Run BloodHound defensively on a schedule.
- Remove unnecessary privileged group membership.
- Limit local administrator rights.
- Review delegated ACLs and inherited permissions.
- Monitor privileged group changes and path-creating permissions.

7. GitHub Redaction and Publication Guidance

- Publish only redacted evidence. Do not upload raw hashes, password material, ticket files, DCSync output, or secretsdump text dumps.
- Keep this report under 03-ad-network/reports/methodology/ and keep raw tool output in a private working folder unless fully sanitized.
- Use placeholders such as , , , and in public documentation.
- Confirm the screenshot file name in the README matches the repository file name exactly, including case, hyphens, and extension.

Do Not Upload

```
*.hash
*_hashes.txt
responder logs with raw NetNTLMv2
kerberoast_hashes.txt
secretsdump output
DCSync dumps
cracked-password files
unredacted BloodHound collection ZIPs if they expose sensitive names
raw command history containing passwords
```

8. Conclusion

The **BloodHound Attack Path Analysis** phase demonstrates a structured and evidence-based AD testing workflow. The report explains the purpose of the test, shows sanitized evidence, connects the activity to real-world risk, and

provides practical remediation guidance. This makes it suitable for a professional cybersecurity GitHub portfolio.